An Introduction to Cluster Creation and Deployment in Kubernetes

Kubernetes automates much of the work of managing containers at scale. It is an open source platform for automated deployment, scaling and management of containerised applications and workloads. It is designed for large deployments. Developed by Google, it is now maintained by the Cloud Native Computing Foundation. In this article, we present an overview of Kubernetes and its deployment.

As technologies like Docker and containerisation have become indispensable parts of developer and operations toolkits and have gained traction in organisations of all sizes, there is a need for better management tools and deployment environments. Kubernetes is one such tool.

What is Kubernetes?

Kubernetes (K8s) is an open source system for automating deployment, scaling, and management of containerised applications. It is the most popular and feature-rich container orchestration system in the world, and offers a wide range of features like:

- Automated rollouts and rollbacks
- Service discovery and load balancing
- Secret and configuration management
- Storage orchestration
- Automatic bin packing
- Batch execution
- Horizontal scaling

Kubernetes is designed for extensibility.

How does Kubernetes work?

Kubernetes operates on a three phase loop of checks and balances.

- **Observe**

  During the observe phase, Kubernetes aggregates a snapshot of the current state of the cluster. Kubelets collect state information about their respective nodes and feed this state back to the main process, giving the main a holistic view of the clusters' current state.